HOW TO USE THE BOOK ?

Scope of Physics
- Awareness on higher learning - courses, institutions and required competitive exams
- Financial assistance possible to help students to climb academic ladder

Learning Objectives:
- Overview of the unit
- Gives clarity on the goals and objective of the topics



- Additional facts related to the topics covered to facilitate curiosity driven learning

Example problems
- To ensure understanding, problems/illustrations are given at every stage before advancing to next level



- Visual representation of concepts with illustrations
- Videos, animations, and tutorials



ICT
- To harness the digital skills to class room learning and experimenting

Summary
- Recap of salient points of the lesson

Concept Map
- Schematic outline of salient learning of the unit

Evaluation
- Evaluate students' understanding and get them acquainted with the application of physical concepts to numerical and conceptual questions

Books for Reference
- List of relevant books for further reading

Practical
- List of practical and the description of each is appended for easy access.

Glossary
- Scientific terms frequently used with their Tamil equivalents

Content Focus
- Sequential understanding of the stationary charges, moving charges, electric current, magnetism and the interlink between electric and magnetic phenomena.

Back Wrapper: NIKOLA TESLA a Serbian-American Engineer (10 July 1856 – 7 January 1943)

Nikola Tesla made breakthroughs in the production, transmission and application of electric power. He invented the first alternating current (AC) motor and developed AC generation and transmission technology. In 1884, he was hired by Edison (discoverer of DC dynamos) later on Tesla became his competitor in this field.

Tesla conducted a range of experiments with mechanical oscillators/generators, electrical discharge tubes, and early X – ray imaging. Tesla in 1890 itself conducted research for wireless lighting and worldwide wireless electric power distribution in his high – voltage, high-frequency power experiments in New York. Unfortunately Tesla could not put his ideas in practical use due to lack of funds.

XII_Physics_First 8 pages Higher.indd 4 14-02-2020 19:30:10